Cultivator

Future of Shrimp Farm

Partner with Cultivator to support next-generation smart aquaculture technology

國立清華大學
NATIONAL TSING HUA UNIVERSITY

NTHU GARAGE
清大創業車庫

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Contents

Problems with aerators

Failures go undetected

1

*Aerator Failures are often caught too late,
resulting in immediate effect of the shrimps stress level and immune system*

Electricity Cost Too High!

2

*The second highest cost goes to the electricity cost,
ranking second to feeding*

Smaller Shrimp

3

*Farmers are forced to harvest their shrimp earlier,
meaning a smaller harvest; lower revenue*

Our Solution

Real-time aerator health monitoring for aquaculture systems

Hardware:

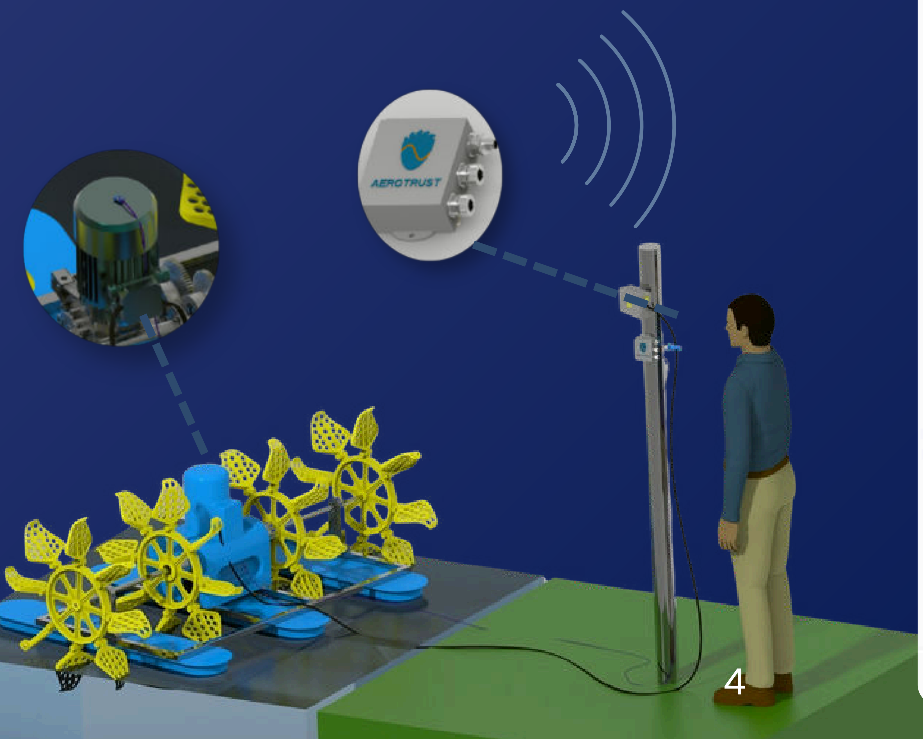
Installed directly on existing paddlewheel aerators to continuously monitor machine health in real time.

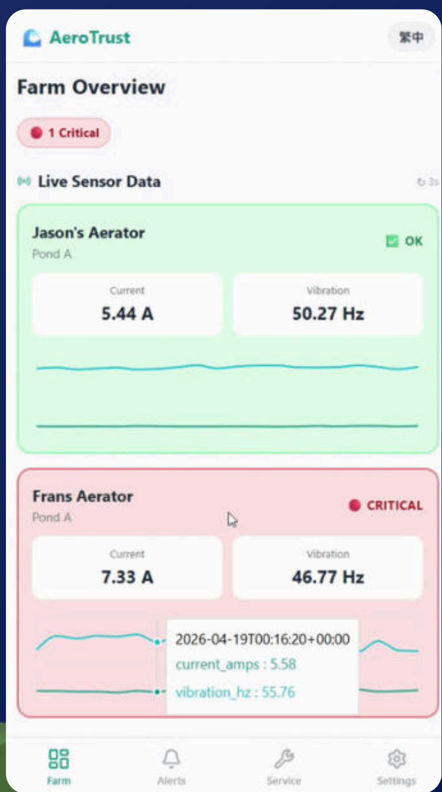
Electrical current	Detect power loss, overload, or motor interruption
Mechanical vibration	Detect abnormal movement, wear, or mechanical failure

Software:

Monitoring & Alert Platform

- Real-time anomaly detection
- Instant fault notifications
- Historical operational analytics





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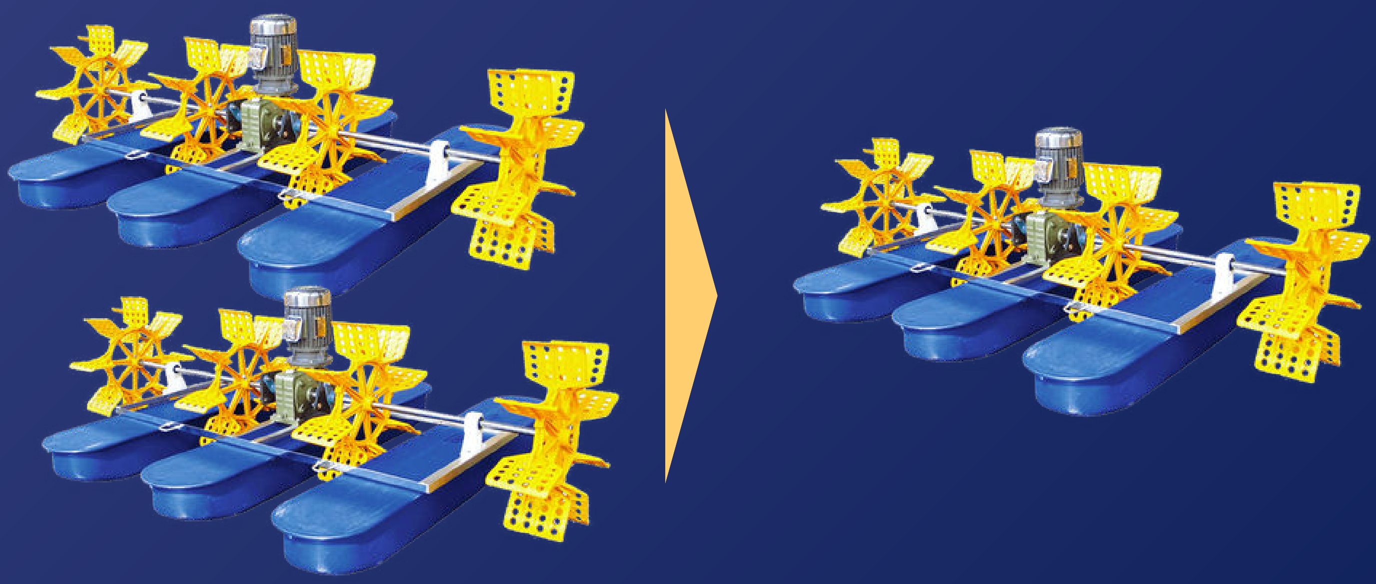
Impact

Aerators are overused. With our product, we will reduce the number of aerators. By just monitoring the aerators, we will be able to see many different impacts!



Save more electricity!

1

By having less aerators, we can cut down the cost of electricity

Reducing Risk

2

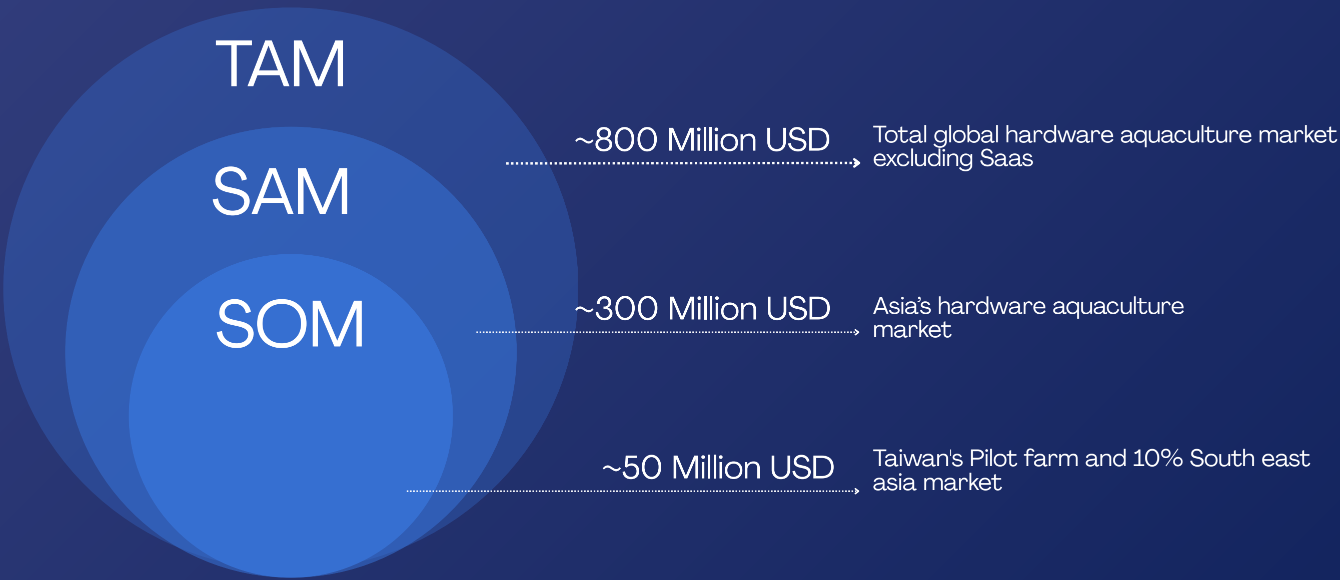
Shrimps can finally live stress free, no more worry about oxygen!

Consistent Shrimp Growth

3

Shrimps can finally live stress free, no more worry about oxygen!

Market Opportunity



Size of the shrimp farming industry

The Taiwan shrimp market reached a value of approximately USD 507 million in 2024, with local production accounting for about 14,285 tons as the industry increasingly shifts toward high-tech, indoor facility-based farming.

Numbers of farms, aerators in use

About 12,000 farms are in shrimp production in Taiwan. Most of them are imported from other countries. Using up to 4 to 8 paddlewheel aerators per pond. (0.1 to 0.3 hectares).



Competitions

Hult Prize

Hult Prize is a global social entrepreneurship competition focused on building startups that solve urgent real-world problems.

For us, this means our project is not only judged as a technical idea, but also as a business that can create social and economic impact.

We won the Hult Prize Nationals

Startup World Cup

We are also joining Startup World Cup Hsinchu, where we will compete with research teams, early-stage startups, and technology-based ventures. This gives us exposure to investors, startup mentors, and industry partners.

STARTUP WORLD CUP

PEGASUS TECH VENTURES

Our team has already won the national stage of the Hult Prize and is now entering the Digital Incubator.

We are also preparing for Startup World Cup Hsinchu, where we will compete with research teams and real startups. Sponsoring us means supporting a student-led innovation team representing Taiwan's startup ecosystem in both social entrepreneurship and technology entrepreneurship platforms.

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Traction

5 farmers approached
3 Commited for Pilot

2 of them currently in Indonesia. They both love our idea, and is willing to adopt it as soon as we get our MVP. The other one is at Taiwan, which we will be working closely with.

On-site Shrimp-farm visit Thailand

Picture with CEO of 886 Apex Aquaculture and Professor Yu Wei

What we're currently doing

We've messaged and will be getting close contact with National Taiwan Ocean University, validating our ideas from there. We've also reached out to multiple aquaculture farms, getting insights and feedback from them.

NTHU Garage Accelerator

As part of NTHU Garage, our team receives startup support, mentorship, and access to an entrepreneurial ecosystem that helps us move from prototype development toward real pilot deployment.

Pitch and presentation for NTHU Garage

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Roadmap: Next 3 Months

Month 1 (May)	Month 2 (June)	Month 3 (July)
Prototype finalization	Field deployment	Validation & Hult Digital Incubator

Month 1 (May) — Prototype finalization

- Finalize device
- Build pilot units
- Waterproof case
- Confirm farms

Output: pilot-ready devices

Month 2 (June) — Field deployment

- Install devices
- Collect data
- Test alerts
- Gather feedback

Output: real-world validation

Month 3 (July) — Validation & Hult Digital Incubator

- Analyze results
- Improve product
- Prepare for scaling

Output: pilot report + next-step strategy

- **Real farm testing**
- **Product validation**
- **Stronger path to Hult Global Stage**

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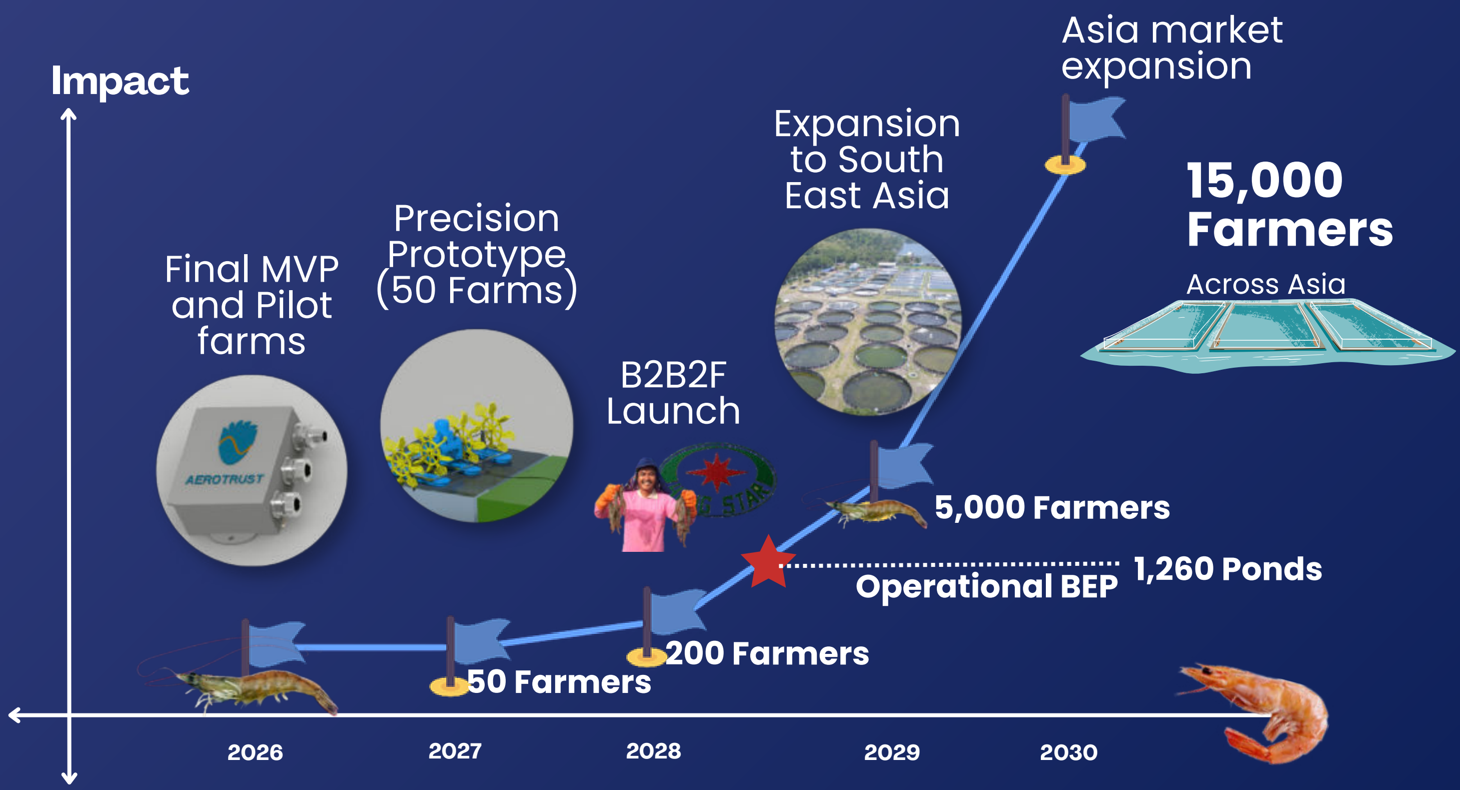
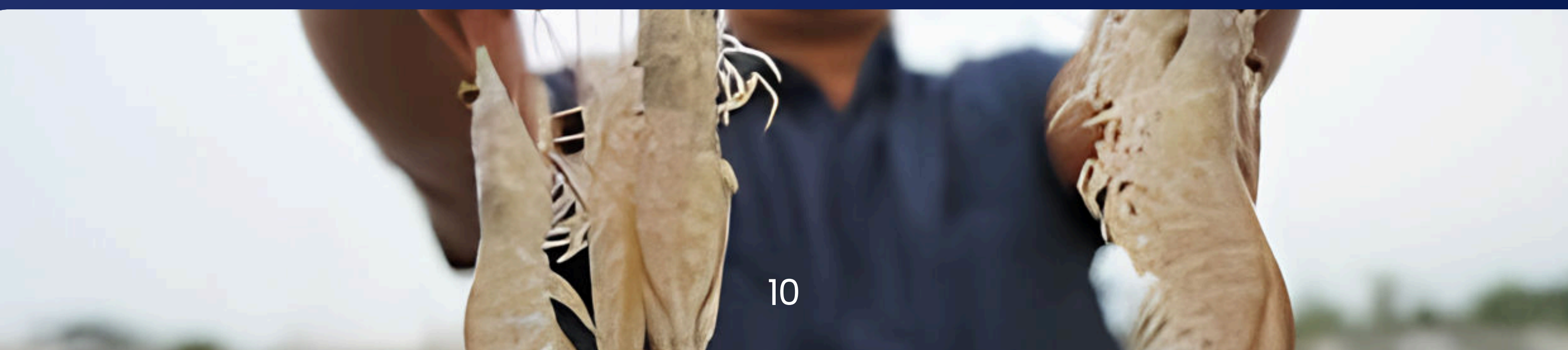
Growth : Next 5 Years

How we plan to grow from student startup to regional aquaculture solution



Long-Term Vision

Build a practical aquaculture monitoring platform for farmers, starting with aerator reliability and growing into smarter farm operations.



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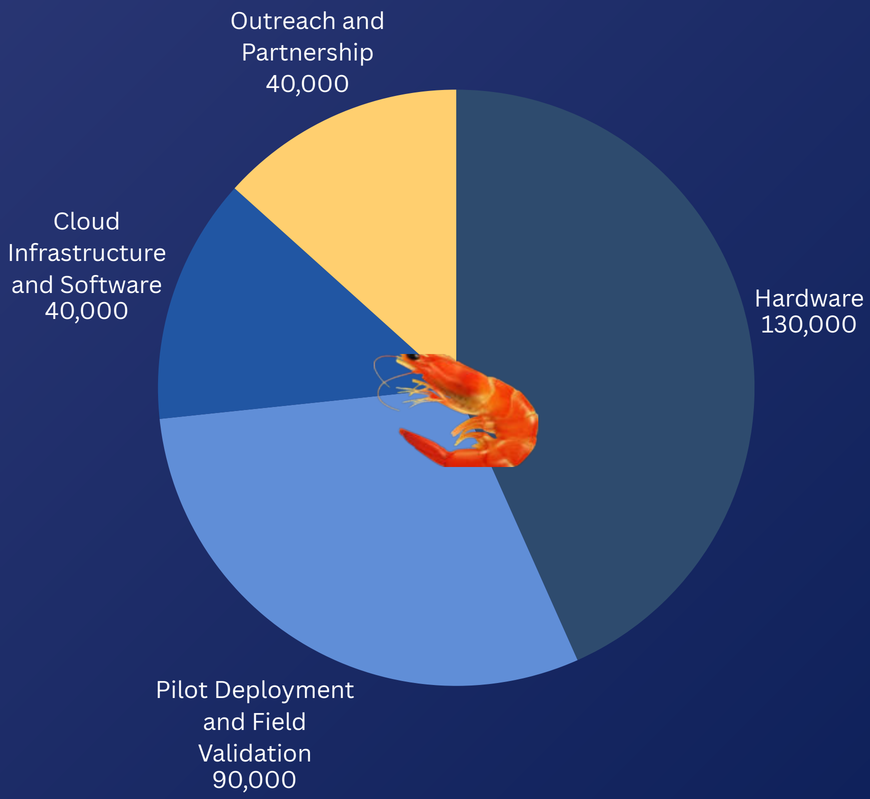
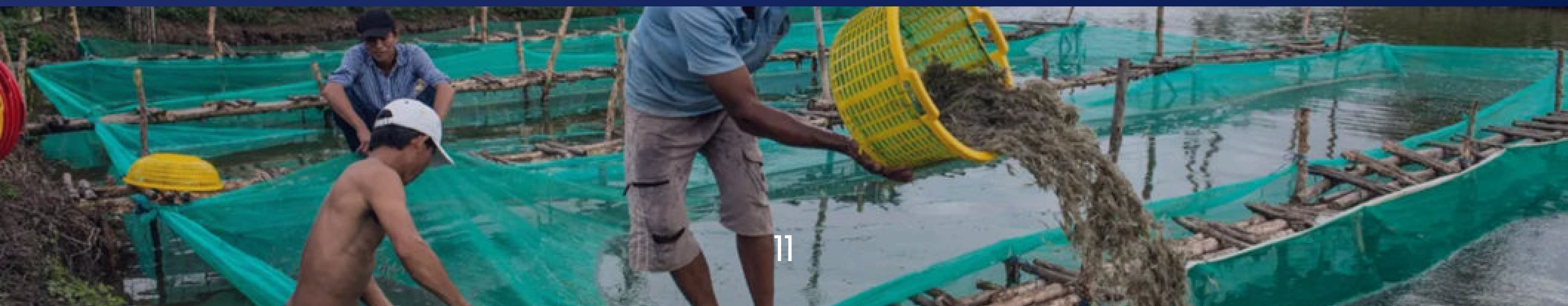
Funding Allocation

Product	**TOTAL = 300,000 NTD**
Hardware Prototyping & sensor Integration	$130,000 NTD
Pilot Deployment & Field Validation	$90,000
Cloud Infrastructure & Software System	$40,000
Outreach & partnership developmeent	$40,000



This funding supports the transition from functional prototype to real-world pilot deployment in aquaculture environments.



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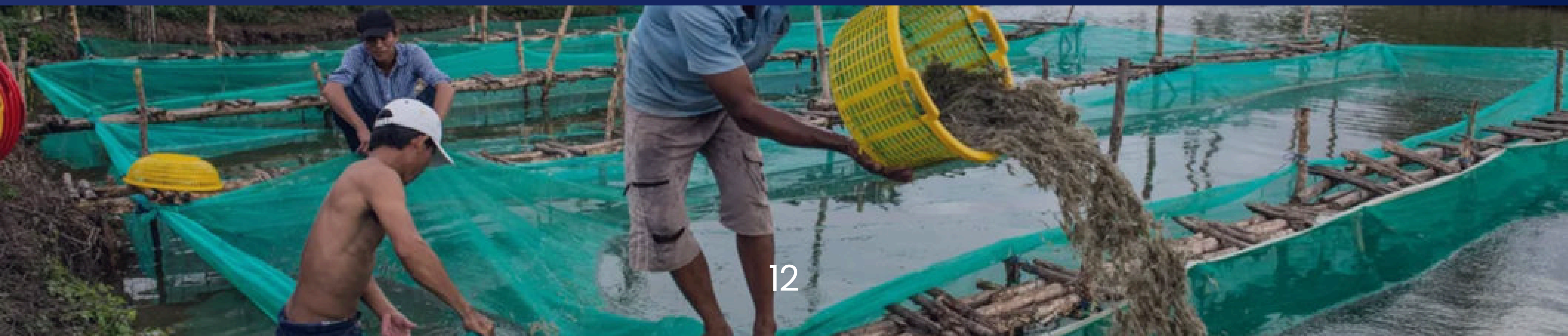
Product and Equipment Needed

Equipment / Item	Purpose
Current sensors	To monitor electrical behavior of the aerator
Vibration sensors	To detect abnormal machine movement
Microcontroller / IoT board	To collect and send data
Waterproof casing	To protect the device in farm conditions
Cables and connectors	To safely attach the device
Communication module	To send alerts from the farm
Power supply / battery components	To keep the device running
Testing tools	To validate safety, signal quality, and reliability
Prototype manufacturing	To build multiple pilot units
Farm installation materials	To deploy devices in real ponds



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Sponsorship Benefits

Cultivator is an early-stage engineering startup developing low-cost monitoring infrastructure for aquaculture systems. Your support will directly accelerate prototype development, pilot deployment, and real-world validation in shrimp farms across Asia.

1 Supporting Partner (10,000 ~ 30,000 NTD)

Sponsor Receives
- Logo on our website
- Logo on selected pitch materials
- Mention in social media posts
- Acknowledgment in our sponsor page
- Short impact report after pilot testing

2 Technical Partner (30,000 ~ 70,000 NTD)

Sponsor Receives
- Logo on website and pitch deck
- Mention during competition-related presentations when appropriate
- Social media feature post
- Sponsor highlight in our progress report
- Invitation to prototype demo
- Access to summarized pilot progress updates

2 Technical Partner (30,000 ~ 70,000 NTD)

Sponsor Receives
- Prominent logo placement on website, deck, and pilot materials
- Dedicated sponsor feature post
- Mention in media, competition materials, and public presentations when appropriate
- Prototype demo session
- Detailed pilot report
- Opportunity to be listed as a strategic supporter
- Priority discussion for future collaboration

We are also open to non-financial sponsorships, including technical mentorship, manufacturing support, cloud infrastructure, sensing hardware, and deployment opportunities.

What Sponsors Can Offer

1. Hardware Support
Examples:

- Sensors
- IoT boards
- Circuit components
- Waterproof casing
- Communication modules
- Testing equipment
- Power supply components

2. Manufacturing Support
Examples:

- PCB fabrication
- 3D printing
- Casing production

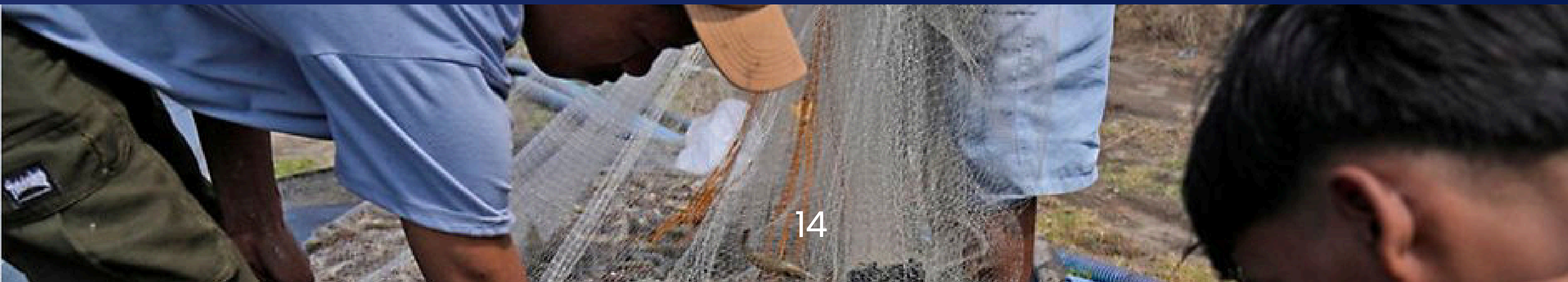
3. Farm or Industry Access
Examples:

- Connecting us with shrimp farmers
- Introducing us to aquaculture companies
- Helping us access pilot sites
- Connecting us with distributors or cooperatives

4. Branding and Media Support
Examples:

- Website support
- Video production
- Pitch deck design
- Public relations
- Social media exposure
- Competition storytelling

This sponsorship is not just supporting a student team. It is helping validate a real technology for aquaculture farmers in Taiwan and Southeast Asia.



Our Team

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